

3-day workshop on modelling lake ecosystems with WET: Tutorial on GOTM-WET setup with QWET

This tutorial covers how to set-up a GOTM-WET model with the QGIS plugin QWET. The material is partly developed by Dr. Dennis Trolle and Dr. Anders Nielsen, WaterITech, during their time at Aarhus University. If you work with SWAT+ and/or GOTM-WET have a look at the [Resource section](#) on their website, where they offer guides and datasets to enable model applications for any location in the world.

Download instructions

Download and install QGIS

We recommend using QGIS3 64 bit newest long-term stable release version (available here: www.qgis.org)

Download and install QWET

You can find and download the latest version of the QWET installer here:

<https://u.pcloud.link/publink/show?code=XZmiT30ZoYEwVCHw4Lu2u1lwUqExT7wdT1Dk>

To enable QWET in QGIS, select Plugins in the top menu in QGIS and select “Manage and Install Plugins”. Then, a pop-up window will appear, find QWET and tick of the box next to QWET. Now a QWET logo will appear QGIS panel. Enabling QWET is also demonstrated in the video tutorials.

QWET procedure

QWET was developed to ease the configuration and application of GOTM-WET models. In QWET users can:

- Setup a GOTM-WET model from scratch
- Manually calibrate and evaluate GOTM-WET models
- Design and execute scenarios

In this three-day workshop, we will setup a GOTM-WET model for Shahe Reservoir via QWET and use the model configuration to apply auto-calibration and sensitivity analysis with parsac. Video tutorials for the Shahe Reservoir are provided in the *Tutorial* folder.

Previous video tutorials with an older version of WET and QWET for the Danish deep Lake Ravn are available [here](#) (developed by Dr. Anders Nielsen).

To configure a GOTM-WET model, users will need information on lake bathymetry, weather, water and nutrient loads as well as food web configuration. QWET allows users to specify rough estimates of these data or specify detailed files (see section below on *Preparing user-defined input files for QWET*). For the Shahe case, all data and files have been prepared in advance and can be found in *Materials* folder on the [WET workshop page](#) (under *Shahe_data*).

Preparing user-defined input files for QWET (optional)

In this section, you can learn how to prepare and use user-defined input files for meteorological data, water and nutrient loads. In addition to these model input files, users can also prepare and load observation files (e.g., temperature, nutrient and oxygen profiles, etc.).

To read in user defined inputs, the following procedure must be followed:

1. Prepare user input files in correct format
2. Create QWET project
3. Place user input files in specific subdirectory of your QWET project
4. Select “User-defined” input in QWET
 - this can currently be done in step 4.1 (weather forcing) and step 4.4 (water and nutrient load)

To be able to read user defined data from external input files, four individual file types may currently be prepared for QWET:

1. meteo_file.dat (meteorological data)
2. inflow.dat (inflow of water)
3. inflow_chem.dat (nutrient concentrations in the inflowing water)
4. *.obs (files with observation data)

Once the files have been created, you should create a new WET project in QGIS. This will generate a QWET working directory with `..\CLIMATE\` and `..\INFLOWS_OUTFLOWS\` subdirectories. Once the QWET project has been created, and you receive the message “The model directory was successfully created”, you should navigate to these subdirectories and place the custom user files here before continuing the model setup in QWET.

Placing meteo_file.dat in correct folder:

Your *meteo_file.dat* should be saved to a “User_xxxxx” folder, which you should create within the “CLIMATE” subdirectory of your QWET projects’ working directory, where xxxxx may be any text string (without spaces). For example: `\QWET_projects\lake_ravn\CLIMATE\User_ravn\meteo_file.dat`

Placing inflow.dat and inflow_chem.dat in correct folder:

Your *inflow.dat* and *inflow_chem.dat* files should be saved to a “User_xxxxx” folder, which you should create within the “INFLOWS_OUTFLOWS” subdirectory of your QWET projects’ working directory, where xxxxx may be any text string (without spaces). For example:

`\QWET_projects\lake_ravn\INFLOWS_OUTFLOWS\User_ravn\inflow.dat`

`\QWET_projects\lake_ravn\INFLOWS_OUTFLOWS\User_ravn\inflow_chem.dat`

Placing *.obs files in correct folder:

Your observation files (*.obs) should be saved to the “observations” folder, which you will find within the “LAKEMODEL\Default” subdirectory of your QWET projects’ working directory. For example: `\QWET_projects\lake_ravn\LAKEMODEL\Default\observations\*.obs`

1. Preparing meteo_file.dat

The meteorological data file comprises eight columns (and rows at preferably subdaily temporal resolution), including (in this order):

1. Date [yyyy-mm-dd]
2. Time [hh:mm:ss]
3. Eastward component (U10) of wind at 10m height [m/s]
4. Northward component (V10) of wind at 10m height [m/s]
5. Air pressure (atmospheric pressure at surface) [hPa]
6. Air temperature [°C]
7. Dew-point temperature [°C]

Note: by default in QWET, dew-point temperature is used, as “hum_method” (under the “Parameters” menu in QWET).

Alternative 1: relative humidity (%) may be used instead (ranging 0-100), whereby the user must change “hum_method” to 1 (accessed under the “Parameters” menu in QWET).

Alternative 2: wet bulb temperature (°C) may be used instead, whereby the user must change “hum_method” to 2 (accessed under the “Parameters” menu in QWET).

8. Cloud cover fraction (ranging 0-1) [-]

The file must be saved in *tab* or *space* delimited ascii (text) format and have the filename and extension “meteo_file.dat”. Note the fixed format of the date (yyyy-mm-dd) and time (hh:mm:ss), which must be followed. If a file is prepared in MS Excel, note that MS Excel may change the date and time formats (each single time the file is opened in excel), and users may therefore need to change the date and time formats in excel just before saving file in a *tab* or *space* delimited text file. An example of a *meteo_file.dat* file of correct format can be found [here](#).

2. Preparing inflow.dat

In QWET, the inflow data file currently represents the aggregated inflow for the entire system. Hence, if data from multiple inflows exist, these should be aggregated into a total inflow value. Note that the model will automatically interpolate (by linear interpolation) between inflow values listed in the file. This means that data in the inflow file do not need to be equidistant (i.e. the time step may vary in the inflow file, for example reflecting actual gauging dates).

The inflow data file comprises three columns including (in this order):

1. Date [yyyy-mm-dd]
2. Time [hh:mm:ss]
3. Discharge [m³/s]

An example of an *inflow.dat* file of correct format can be found in the Shahe data folder. Another example can be found [here](#).

3. Preparing inflow_chem.dat

In WET, the inflow chemistry data file currently represents the aggregated inflow for the entire system. Hence, if nutrient data from multiple inflows exist, these should be aggregated into a total inflow value (i.e. flow-weighted inflow concentrations). Note that the model will automatically interpolate (by linear interpolation) between concentration values listed in the file. This means that data in the inflow chemistry file do not need to be equidistant (i.e. the time step may vary in the inflow file, for example reflecting data and dates of actual sampling dates).

The inflow chemistry data file comprises nine columns including (in this order):

1. Date [yyyy-mm-dd]
2. Time [hh:mm:ss]
3. Particulate organic phosphorus concentration (PPOM) [mg P/L]
4. Dissolved organic phosphorus concentration (PDOM) [mg P/L]
5. Phosphate concentration [mg P/L]
6. Particulate organic nitrogen concentration (NPOM) [mg N/L]
7. Dissolved organic nitrogen concentration (NDOM) [mg N/L]
8. Nitrate concentration [mg N/L]
9. Ammonium concentration [mg N/L]

An example of an *inflow_chem.dat* file of correct format can be found in the Shahe data folder. Another example can be found [here](#).

4. Preparing *.obs files

In WET, you can compare model output with time series observations for a range of different state variables. You may prepare one file for each state variable that you have observations for.

The observation files must comprise four columns including (in this order):

1. Date [yyyy-mm-dd]
2. Time [hh:mm:ss]
3. Depth from surface [m] (Negative values. For example, -1 represents 1 meters depth below surface)

4. Value of state variable concentration (in units corresponding to model output units, e.g. Celsius for temperature, mg/L for nutrients and oxygen, and $\mu\text{g/L}$ for chlorophyll *a*)

An example of an *.obs file of correct format can be found in the Shahe data folder. Another example can be found [here](#). The names of the individual observations files must match that defined in the *wet_obs.xml* file that is also located in the "observations" folder. For further details, please watch this [tutorial video](#) on how to format and work with observations in WET.